Form [SfDS]

* Derivative Rules

$(u+v)' = u' + v'$

$(u\,v)' = u'\,v + u\,v'$

$\left(\frac{u}{v}\right)' = \frac{u'\,v - u\,v'}{v^2}$

$(u^n)' = n\,u^{n-1}\,u' \quad (n \in \mathbb{R})$

$(e^u)' = u'\,e^u$

$(a^u)' = u'\,a^u \ln a \quad (a \in \mathbb{R}^+ \backslash \{1\})$

$(\ln u)' = \frac{u'}{u}$

$(\log_a u)' = \frac{u'}{u \ln a} \quad (a \in \mathbb{R}^+ \backslash \{1\})$

* Primitives

$\int \sin^2 x\,dx = \frac{1}{2}x - \frac{\sin 2x}{4}$

$\int \cos^2 x\,dx = \frac{1}{2}x + \frac{\sin 2x}{4}$

f(x)	∫ f(x) dx
k	$kx + C, \quad k \in \mathbb{R}$
$u^n u'$	$\frac{u^{n+1}}{n+1} + C, \quad n \in \mathbb{R}\backslash\{-1\}$
$\frac{u'}{u}$	$\ln\lvert u \rvert + C$
$e^u u'$	$e^u + C$
$a^u u'$	$\frac{a^u}{\ln(a)} + C, \quad a \in \mathbb{R}^+\backslash\{1\}$
$\frac{u'}{1+u^2}$	$\arctan(u) + C$
$\frac{u'}{\sqrt{1-u^2}}$	$\arcsin(u) + C$

Log Properties

$\log(a \cdot b) = \log(a) + \log(b)$

$\log(a^b) = b \cdot \log(a)$

$\ln(e^a) = a$

$\log\left(\frac{a}{b}\right) = \log(a) - \log(b)$

$\log(1) = 0$

$\log_a(a) = 1$

$a^{\log_a(k)} = k$

$\log_b(m) = \frac{\log(m)}{\log(b)}$

1. Statistical Distributions

Definitions

Units → The basic objects on which the data is collected.

Variable → Characteristic of units that can take on different values (in other words, something that can vary).

Categorical variables refer to variables with values that can't be quantifiable.
- **Nominal**, which describes a name, label or category without a logical order (e.g., gender);
- **Ordinal**, whose values are defined by an order between the different categories (e.g., education level).

Quantitative variables have numerical values with magnitudes that can be placed in a meaningful order with consistent intervals, also known as numerical.
- **Discrete**, which means that can only take on a set number of values (e.g., only whole numbers);
- **Continuous**, which means that can take on any value and any value between values (e.g., out to an infinite number of decimal places).

o **Central tendency measures** describe the "center" around which the data is distributed. (*Mean* $\bar{x}/\mu$, *Median - better for outliers*)
o **Variability measures** describe "data spread" or how far away the measurements are from the center. (*Variance* and *Standard Deviation*)
o **Relative Standing measures** describe the relative position of specific measurements in the data. (*Percentile* & *Quartiles*)

Correlation (r/ρ): describe the strength and direction of association between 2 quantitative variables [is unit free; the x and y variables do NOT need to be on the same scale]
- **Ranges** between -1 and 1 (0 means no linear relationship). $r = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = \frac{E[XY] - E[X] \times E[Y]}{\sqrt{Var(X) \times Var(Y)}}$

Random Variable is a variable that can take on certain numerical values with certain probabilities.
- The collection of these probabilities is called the **probability distribution** for the random variable.
- A probability distribution specifies how the total probability (which is always 1) is distributed among the various possible outcomes.

Random Variable Parameters

Expected Value of X (μ_X) $E[X] = \mu_X = \sum_{i=1}^n x_i f(x_i)$ $E[X] = \int_{-\infty}^{+\infty} x f(x) dx$

Variance of X (σ_X^2) → Standard Deviation of X (σ_X) $Var[X] = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x) dx$

DISCRETE (Finite or Infinite Numerable Support)

♦ Probability Mass (Function) → $f(x) = P[X = x]$

Distribution of Probability (Function) → $F(x) = P[X \leq x] = \sum_{x_k \leq x} f(x_k)$

1. Binomial Distribution

$X \sim Bin(n,p)$ (Number of successes in n Bern trial/experiment)

$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, \ldots, n$ and $\binom{n}{x} = \frac{n!}{x!\,(n-x)!}$ $\quad E[X] = np \quad Var[X] = np(1-p)$

- Is frequently used to model the number of successes in a sample of size n drawn with replacement from a population of size N.
- The formula can be understood as follows: *x successes occur with probability p^x and $n - x$ failures occur with probability $(1-p)^{n-x}$*
  → However, the x successes can occur anywhere among the n trials, and there are $\binom{n}{x}$ different ways of distributing x successes in a sequence of n trials.

2. Bernoulli Distribution (Binomial's $n = 1$) $X \sim Ber(p)$ (Coin)

Bern Proofs A (1) - Success (p) [$P(X=1) = p = 1 - P(X=0) = 1-q$] $\quad \bar{A}$ (0) - Failure (q) ($= 1-p$)

$f(x) = p^x (1-p)^{1-x}, \quad x = 0, 1 \quad E[X] = p \quad Var[X] = p(1-p) = p \times q$

- It can be used to represent a (possibly biased) coin toss where 1 and 0 would represent heads and tails (or vice versa), respectively, and p would be the probability of the coin landing on heads or tails, respectively. → In particular, **unfair coins** would have $p \neq 0.5$

3. Poisson Distribution

$X \sim Po(\lambda)$ (Counting in Continuous Ranges)

$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0,1,2,\ldots \quad e = 2.71828\ldots$) (Euler's number) $\quad \lambda \in \mathbb{R}^+ \quad E[X] = Var[X] = \lambda$

CONTINUOUS (Continuous Support)

~~Probability~~ Function ~~(Mass)~~ → Makes no sense! $P[X = x] = 0$

Probability Density Function $f(x)$ **p.d.f.** Probability of assuming a **specific value x**

Cumulative Distribution Function $F(x)$ **c.d.f.**

$f(x) \geq 0 \quad \int_{-\infty}^{+\infty} f(x)\,dx = 1 \quad F(x) = P[X \leq x] = \int_{-\infty}^{x} f(x) dx$, for all x

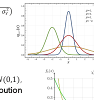
```
rxxx(n,…)               → Geração de n observações aleatórias com a distribuição especificada nos parâmetros
dxxx(x,)                → Função de probabilidade (num ponto x) - f(x)
pxxx(p,…, lower.tail=)  → Função Distribuição F(x) = P[X ≤ x]
qxxx(q,…, lower.tail=)  → Dada uma probabilidade, determina o (primeiro) x em que esse acumulado é atingido (quantil)

binom(size, prob)             → Binomial      lower.tail = FALSE  P[X > x]  Pq são v.a. DISCRETAS
pois(lambda)                  → Poisson
norm(mean,sd)                 → Normal        lower.tail = FALSE  P[X ≥ x] = P[X > x]  Pq são v.a. CONTÍNUAS
t(graus_de_liberdade)         → t-Student
chisq(df)                     → Chi-Squared
f(df1,df2)                    → F
```

1. Normal Distribution

$X \sim N(\mu, \sigma^2) \quad f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

$Z = \frac{X-\mu}{\sigma} \sim N(0,1)$ (z-score) $\quad$ TAN $(X \pm Y) \sim N(\mu_X \pm \mu_Y, \sqrt{\sigma_X^2 + \sigma_Y^2})$

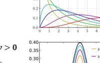
TLC $\bar{X} = \frac{\sum_{i=1}^n x_i}{n} \sim N\left(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}\right)$



2. Chi-Squared Distribution

If $X_1, X_2, \ldots, X_n$ are n independent random variables $X \sim N(0,1)$, then the following quantity follows a **Chi-squared** distribution with n **degrees of freedom**. $\quad E[X] = n \quad Var[X] = 2n$

$V = X_1^2 + X_2^2 + \ldots + X_n^2 \sim \chi^2_{(n)}, \; n > 0 \qquad X \sim \chi^2_{(n)}$



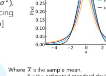
3. Student's t Distribution

$X \sim t_{(v)} \quad v > 0$

The *t-distribution* is **symmetric** and **bell-shaped**, like $N(\mu, \sigma^2)$, but has heavier tails, meaning that it is more prone to producing values that fall far from its mean. [v → **degrees of freedom**]

Média: $\mu = 0$ para $k > 1$

Variância: $\sigma^2 = \frac{k}{k-2}$ para $k > 2$



n>30 → T-Student Test Statistic $t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{(n-1)}$

Where $\bar{X}$ is the sample mean, $\hat{\sigma}$ is the estimated standard deviation, μ_0 is the hypothesized population mean, n is the sample size.

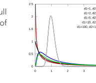
4. Snedcor's F Distribution

$X \sim F(v_1, v_2)$

Fisher-Snedecor distribution that arises frequently as the null distribution of a test statistic, most notably in the analysis of variance (**ANOVA**), e.g., F-test.

Média: $\mu = \frac{d_2}{d_2 - 2}, \quad d_2 > 2$

Variância: $\sigma^2 = \frac{2d_2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}, \quad d_2 > 4$



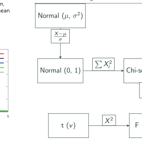
Properties of Expected Value $E[X]$
- Being X and Y two v.a's and k, a and b constants $\mathbb{R}$
  - $E[k] = k$
  - $E[kX] = kE[X]$
  - $E[X \pm Y] = E[X] \pm E[Y]$
  - $E[aX + bY] = aE[X] + bE[Y]$
  - $E[XY] = E[X] \times E[Y]$, if X and Y are independent

Properties of $Var[X]$
- $Var(k) = 0$
- $Var(X + k) = Var(X)$
- $Var(kX) = k^2 Var(X)$
- $Var(X \pm Y) = Var(X) + Var(Y) \pm 2Cov(X,Y)$
- $Cov(aX, bY) = ab\,Cov(X,Y)$

$Var[X] = E[(X - E[X])^2] = E[(X - \mu_X)^2] = E[X^2] - (E[X])^2 = \sigma_X^2$

♦ If X and Y are **independent**, then $Cov(X,Y) = 0$

♦ If X it is a v.a. of mean μ and variance σ^2. So v.a. has $W = \frac{(X-\mu)}{\sigma}$ $E[W] = 0$ and $Var(W) = 1$



2. Point Estimation | Confidence Intervals | Hypothesis Test.

Definitions

Let $X_1, \ldots, X_n$ be n random variables (R.V.) arising from a random sample. The corresponding observed values of a specific **R.V.** are denoted: $x_1, \ldots, x_n$.
- The range of possible values of the parameter θ is called the **Parameter Space** Ω. F.e., If μ denotes the mean grade of all university students, then the parameter space (assuming a 20-point grading scale): $\Omega = \{\mu: 0 \leq \mu \leq 20\}$
- The function $X_1, X_2, \ldots, X_n$, that is, the statistic $T(X_1, X_2, \ldots, X_n)$, used to estimate θ is called a **Point Estimator** of θ. F.e., the function: $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$ is a point estimator of the population mean μ.
- The function $T(x_1, x_2, \ldots, x_n)$ computed from a set of data is an observed **Point Estimate** of θ. F.e., If x_i are the observed grade point averages of a sample of 88 students, then: $\bar{x} = \frac{1}{88}\sum_{i=1}^{88} x_i = 14.5$ is a point estimate of μ.

Method of Moments

1. $E(X^k)$ is the k^{th} theoretical moment of the distribution (about the origin), for $k = 1, 2, \ldots$
2. $m_k = \frac{1}{n}\sum_{i=1}^n X_i^k$ is the k^{th} sample moment, for $k = 1, 2, \ldots$

Method of moments estimators are found by equating the first k sample moments to the corresponding k population moments, until we have as many equations as we have parameters. Then, we solve the resulting system of simultaneous equations. More precisely, define

$$m_1 = \frac{1}{n}\sum_{i=1}^n X_i^1, \mu_1' = E[X^1] \qquad m_2 = \frac{1}{n}\sum_{i=1}^n X_i^2, \mu_2' = E[X^2] \qquad \ldots \qquad m_k = \frac{1}{n}\sum_{i=1}^n X_i^k, \mu_k' = E[X^k]$$

The method of moment estimator is obtained by solving the following system of equations: $m_1 = \mu_1', \ldots, m_k = \mu_k'$

Method of Moments - Example Bernoulli Distribution

Suppose $X_1, \ldots, X_n$ are i.i.d. that follow a Bernoulli with parameter p. The first theoretical moment is: $E(X_i) = p$ (1)
We have just one parameter for which we are trying to derive the method of moments estimator. Therefore, we need just one equation.
Equating the first theoretical moment about the origin with the corresponding sample moment we get: $p = \frac{1}{n}\sum_{i=1}^n X_i$ (2)

Method of Moments - Example Normal Distribution

Suppose now that $X_1, \ldots, X_n$ are i.i.d. that follow a Normal(μ, σ^2). We have: $m_1 = \bar{X} \mid m_2 = (1/n)\sum X_i^2$ $\quad \mu_1' = E[X^1] = \mu \mid \mu_2' = E[X^2] = \mu^2 + \sigma^2$
Thus, we must solve: $\frac{1}{n}\sum X_i = \mu$ and $\frac{1}{n}\sum X_i^2 = \mu^2 + \sigma^2$ | Solving for μ and σ^2 yields the method of moments estimator $\hat{\mu} = \bar{X}$ | $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$

Maximum Likelihood Estimator (MLE)

- Suppose we have a random sample $X_1, X_2, \ldots, X_n$, whose assumed probability distribution depends on some unknown parameter θ.
- Our primary goal here will be to find a point estimator, such that is produces a "good" point estimate of θ.
- **F.e.,** if we plan to take a random sample for which X_i are assumed to be normally distributed with mean μ and variance σ^2, then our goal will be to find a good estimate of μ, using the data $x_1, x_2, \ldots, x_n$, that we obtained from our specific random sample.

It seems reasonable that a good estimate of the unknown parameter θ would be the value of θ that **maximizes the probability** (i.e., likelihood) of getting the data we observed.

If $X_1, \ldots, X_n$ are an **i.i.d.** sample from a population with pdf $f(x|\theta_1, \ldots, \theta_k)$, the **likelihood function** is defined by

$$L(\theta|x) = L(\theta_1, \ldots, \theta_k | x_1, \ldots, x_n) = \prod_{i=1}^n f(x_i|\theta_1, \ldots, \theta_k)$$

- **Typically**, we take the log of the likelihood and obtain the log-likelihood function $logL(\theta|x) = \sum_{i=1}^n log(f(x_i|\theta_1, \ldots, \theta_k))$
  - Since the logarithm is a strictly increasing function, maximizing the log of the likelihood is the same as maximizing the likelihood.
  - Then, possible candidates for the MLE are the values of $(\theta_1, \ldots, \theta_k)$ that solve $\frac{\partial}{\partial \theta_i} logL(\theta|\mathbf{X}) = 0, i = 1, \ldots, k$

Putting our calculus hats on, it is easier to differentiate the log-likelihood function than the likelihood!

MLE - Bernoulli Distribution

Suppose we have a random sample $X_1, X_2, \ldots, X_n$ where $X_i = 0$ if a randomly selected student does not own a sports car, and $X_i = 1$ if a r.s. student does own a sports car.
If the X_i are independent Bernoulli random variables with unknown parameter p, then the probability mass function of each X_i is: $f(x_i|p) = p^{x_i}(1-p)^{1-x_i}$
Then, the likelihood function is, by definition: $L(p|x_1, \ldots, x_n) = \prod_{i=1}^n f(x_i|p) = p^{x_1}(1-p)^{1-x_1} \times p^{x_2}(1-p)^{1-x_2} \times \ldots \times p^{x_n}(1-p)^{1-x_n}$
And the log-likelihood function is: $logL(p|x_1, \ldots, x_n) = (\sum_{i=1}^n x_i)log(p) + (n - \sum_{i=1}^n x_i)log(1-p)$
Now, taking the **derivative** of the log-likelihood, and setting it to 0, we get: $\frac{\partial logL(p)}{\partial p} = \frac{\sum_{i=1}^n x_i}{p} - \frac{(n - \sum_{i=1}^n x_i)}{1-p} = 0$
Multiplying through by $p(1-p)$, we get: $(\sum_{i=1}^n x_i)(1-p) - (n - \sum_{i=1}^n x_i)p = 0$
Solving for p will yield the MLE estimator: $\hat{p} = \frac{\sum_{i=1}^n x_i}{n}$

MLE - Poisson Distribution

Consider the population density function of a Poisson distribution: $f(x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$ and the following sample: (95,100,80,70,110,98,97,90,70). Obtain the **ML** point estimator and point estimate for μ.

The likelihood function for a Poisson distribution is given by: $L(\lambda|x) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$
The log-likelihood function is given by: $ln(L(\lambda|x)) = \ldots = n\bar{x} - n\lambda - \sum_{i=1}^n \ln(x_i!) = n\bar{x} - n\lambda - C$ where C is a constant that does not depend on λ and $\bar{x}$ is the sample mean.
The log-likelihood function can be **maximized** by maximizing the term $n\bar{x}$ and **minimizing** the term $n\lambda$. The **MLE**stimator is obtained by **maximizing** the l-l.
The maximum likelihood estimator is obtained by solving the equation: $\frac{\partial ln(L(\lambda|x))}{\partial \lambda} = 0$
The 1st derivative of the log-likelihood function is given by: $\frac{\partial ln(L(\lambda|x))}{\partial \lambda} = \frac{\partial}{\partial \lambda}(n\bar{x} - \lambda - C) = -n + \frac{1}{\lambda}\sum_{i=1}^n x_i$
Setting the derivative to 0 to solve for λ: $-n + \frac{1}{\lambda}\sum_{i=1}^n x_i = 0 \Leftrightarrow \frac{1}{\lambda}\sum_{i=1}^n x_i = n \Leftrightarrow \lambda = \frac{\sum_{i=1}^n x_i}{n}$
Since λ is the parameter of the Poisson distribution, the maximum likelihood estimator for λ is the sample mean: $\hat{\lambda} = \frac{\sum_{i=1}^n x_i}{n}$
Thus, the maximum likelihood estimator for λ is the mean of the sample. $\quad \therefore \hat{\lambda} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x} = \frac{95+100+80+70+110+98+97+90+70}{9} = 90$

Maximizing the likelihood or the log-likelihood will always yield the same result!

Ideal Properties of an Estimator

- **Unbiasedness**: the expected value of the estimator should be equal to the true value of the parameter we are estimating. $E(T) = \theta, \forall\theta \in \Omega$
- **Efficiency**: the variance of our estimator is the smallest (could be in relative or absolute terms). $MSE(T) = E\{(T-\theta)^2\}$
  The estimator with lower MSE is considered "better", because it contains estimates more concentrated around θ true value.
- **Consistency**: as the sample increases, the estimator will be closer to the true value of the parameter we are estimating. $\lim_{n \to +\infty} E\{(T_n - \theta)^2\} = 0$

Hypothesis Testing

Research question	Is the population mean different from μ_0	Is the population mean greater than μ_0	Is the population mean less than μ_0
Null Hypothesis, H_0	$\mu = \mu_0$	$\mu = \mu_0$	$\mu = \mu_0$
Alternative Hypothesis, H_1	$\mu \neq \mu_0$	$\mu > \mu_0$	$\mu < \mu_0$
Type of Hypothesis test	Two-tailed, non-directional	Right-tailed, directional	Left-tailed directional

μ_0 is the hypothesized population mean

5 Steps

1. Write Hypothesis
- Null Hypothesis (H_0): No effect/ difference.
- Alternative Hypothesis (H_1): Suggestions of an effect or difference

2. Test Statistic (t_{obs})
- Calculate the test statistic from the sample data. (Choose the appropriate test statistic based on the hypothesis and the type of data.) $N > 30$ → T- Student

3. P-value/Critical Region
- Calculate the p-value (*Under the H_0, the P of observing a test statistic as extreme as the one calculated from the sample data*)
  - **Small p-value:** ($p - value \leq \alpha$) Evidence against the null hypothesis. → Suggestions that the observed data is unlikely to have occurred if the null hypothesis is true (Reject H_0)
  - **Large p-value:** ($p - value > \alpha$) Fail to reject the null hypothesis. → The observed data is likely (or not unsual) to have occurred if the null hypothesis is true. (Fail to reject H_0)

α **(Significance Level):** The probability of rejecting the null hypothesis when it is true in the population.
For a **t-test** (testing means), the $p - value$ can be expressed as:

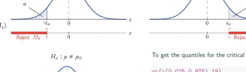
$$p - value = P(\text{Observed Data}|H_0) = P(t_{n-1} \leq t_{obs} \mid H_0 \text{ is true})$$

where t_{n-1} is the t-distribution with $n-1$ degrees of freedom and t_{obs} is the observed t-statistic.
I.e. it's the probability of obtaining a test statistic T more extreme (or as extreme) as the observed test statistic t_{obs}, given that H_0 is true.

- Determine the Critical Region.

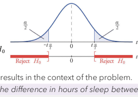
$$|t_{obs}| \geq t_{1-\alpha/2; n-k-1}$$

where $t_{1-\alpha/2;n-k-1}$ corresponds to the quantile of the t-student distribution.
⚠ Notice that we consider $\alpha/2$ because our test is **bilateral** (it has ≠ on H_1).



More Coincidence → Higher $p - value$

4. Decision:
- Compare the p-value with the significance level (α) OR Compare the test statistic with the critical value.
  - If $ET \in CR$ or $p - value \leq \alpha$ → **Reject** H_0
  - If $ET \in RNC$ or $p - value > \alpha$ → **Fail to Reject** H_0



To get the quantiles for the critical region:

```
qt(c(0.025,0.975),19)
## [1] -2.093024  2.093024
```

5. Interpretation:
- Make a decision based on the comparison and interpret the results in the context of the problem.

In conclusion, we can state with a 5% significance level that the difference in hours of sleep between the treated students and the control group is different from zero.

One sample proportion

- First, we should define the hypothesis: $\begin{cases} H_0: p = 0.4 \\ H_1: p > 0.4 \end{cases}$
- The test statistic in this case is given by

$$Z_{obs} = \frac{\hat{p} - 0.4}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim N(0,1)$$

- The critical region for this test: $Z_{obs} \geq Z_{1-\alpha}$ where $Z_{1-\alpha}$ is the $1-\alpha$ percentile of the standard normal distribution

- In R, we can use the function `prop.test()`:

```
prop.test(x=129,
          n=136,
          p=0.4,
          alternative = c('greater'))
```

where x corresponds to the number of 'sucesses' (here the number of patients that had a stroke), n corresponds to the size of the sample, p the probability value we want to test and the alternative specifies the alternative hypothesis.

Other Parameters

Research question	Is the population proportion different from p_0	Is the population proportion greater than p_0	Is the population proportion less than p_0
Null Hypothesis, H_0	$p = p_0$	$p = p_0$	$p = p_0$
Alternative Hypothesis, H_1	$p \neq p_0$	$p > p_0$	$p < p_0$
Type of Hypothesis test	Two-tailed, non-directional	Right-tailed, directional	Left-tailed directional

p_0 is the hypothesized population proportion.

Type I and Type II Errors

Decision/Reality	H_0 is true	H_0 is false
Reject H_0	Type I error (α)	Correct decision
Fail to reject H_0	Correct decision	Type II error (β)

Power of Test

- The probability of rejecting the null hypothesis, given that the null hypothesis is false, is known as power. In other words, power is the probability of correctly rejecting H_0.

$$Power = 1 - \beta$$

where β = probability of committing a Type II Error.
- The power of a test can be increased in a number of ways, for example increasing the sample size, decreasing the standard error, increasing the difference between the sample statistic and the hypothesized parameter, or increasing the alpha level.

Confidence Intervals

$\bar{X} \pm erro = \bar{X} \pm t_{1-\alpha/2}\left(\frac{\hat{\sigma}}{\sqrt{n}}\right)$

When constructing a **CI for a population mean** the **point estimate** is the sample mean, $\bar{X}$.
The multiplier is the percentile $1 - \alpha/2$ from t distribution. And the standard error is equal to $\frac{\hat{\sigma}}{\sqrt{n}}$.

Normal Dist. $]I_{0.95}[_\mu = \bar{X} \pm \frac{1.96}{\sqrt{n}}\sigma$ ou $]I_{0.99}[_\mu = \bar{X} \pm \frac{2.576}{\sqrt{n}}\sigma$

The correct interpretation of a 95% confidence interval is that 'we are 95% confident that the population parameter is between X and X.'

```
# Prepare the space
conf <- 0.95   # Confidence Level q <- qnorm(conf+tail/2)
tail <- 1-conf # weight of tails #qnorm(0.975) = 1.96

# Calculation of the Concrete Interval
media <- mean(bd_oculos_sol$Pub.Mk)       LB <- media-error
dp  <- sd(bd_oculos_sol$Pub.Mk)           UB <- media+error
error <- q*dp/sqrt(nrow(bd_oculos_sol)) round(c(LB,UB),2)
```

Inference for Two samples

Two sample proportion

- First, we should define the hypothesis: $\begin{cases} H_0: p_A = p_B \\ H_1: p_A > p_B \end{cases}$
- The test statistic in this case is given by

$$Z_{obs} = \frac{\hat{p}_A - \hat{p}_B}{\sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{n_A} + \frac{\hat{p}_B(1-\hat{p}_B)}{n_B}}} \sim N(0,1)$$

- The critical region for this test: $Z_{obs} \geq Z_{1-\alpha}$

```
patients <- c(140, 240) | hypertension <- c(119, 200)
prop.test(hypertension, patients, alternative = c('greater'))
```

Two sample mean test

- First, we should define the hypothesis: $\begin{cases} H_0: \mu_A = \mu_B \\ H_1: \mu_A < \mu_B \end{cases}$
- If the samples have different variances, the test statistic in this case is given by

$$t_{obs} = \frac{\hat{\mu}_A - \hat{\mu}_B}{\sqrt{\frac{\hat{\sigma}_A^2}{n_A} + \frac{\hat{\sigma}_B^2}{n_B}}} \sim t(n_A + n_B - 2)$$

- If the samples have the same variance, the test statistic

$$t_{obs} = \frac{\hat{\mu}_A - \hat{\mu}_B}{\hat{\sigma}\sqrt{\frac{1}{n_A} + \frac{1}{n_B}}} \sim t(n_A + n_B - 2)$$

- The critical region for this test: $t_{obs} \leq t_{\alpha; n_A+n_B-2}$

```
t.test(schoolA, schoolB, alternative = c('less'),
       var.equal=TRUE)
```

We define var.equal = TRUE, because we did not reject the null on the previous test.

Test of Variances

- Before we test for the difference in means between the two samples, we must determine whether the variances are equal through a test.

The hypothesis are: $\begin{cases} H_0: \sigma_A^2 = \sigma_B^2 \\ H_1: \sigma_A^2 \neq \sigma_B^2 \end{cases}$

- The test statistic is $F_{obs} = \frac{\hat{\sigma}_A^2}{\hat{\sigma}_B^2} \sim F(n_A - 1, n_B - 1)$
- And, the critical region for this test is: $F_{obs} \geq F_{\alpha; n_A-1, n_B-1}$

```
schoolA = c(38.9, 61.2, 73.3, 51.8, 63.4, 64.6, 58.4, 58.8, 68.5)
schoolB = c(67.8, 60, 63.4, 76, 89.4, 73.3, 67.3, 61.3, 62.4)
var.test(schoolA, schoolB)
```

3. Multiple Linear Regression Model

RA *"aims to give empirical content to relations"*. Specifically, it allows to uncover a relationship between 2 or + variables.

RA | Cross-Sectional Data

- An important feature of CSD: they are obtained by random sampling from the underlying population.
  F.e., suppose that y_i is the ***i*-th observation** of the **dependent variable** and x_i is the ***i*-th observation** of the **explanatory variable**
  - Random sampling means that $\{(y_i, x_i)\}$ is an independent sequence.
  - This implies that for $i \neq j$ → $Cov(y_i, y_j) = Cov(x_i, x_j) = Cov(y_i, x_j) = 0$

y	x
Dependent variable	Independent variable
Explained variable	Explanatory variable
Response variable	Control variable
Predicted variable	Predictor variable
Regressand	Regressor

Causal Effect of x on y: *"How does variable y change if variable x is changed but all other relevant factors are held constant?"* **Ceteris Paribus**
Ceteris Paribus: *"other (relevant) factors being equal"*, plays an important role in causal analysis.

RA | Simple Linear Regression Model $\quad Y = \beta_0 + \beta_1 X + \epsilon$ | ϵ or u

- Assume that we have **have two (observable) variables** y and x. And that we are interested in: *studying how y varies with changes in x*
- u is called the **error term**, random disturbance ("disturbs an otherwise stable relationship"). → Represents **unobserved factors** other than x that affect y.
  Conditional mean independence Assumption → $E[u \mid x] = 0$ → Unobserved factors do not contain information on explanatory variables.

Deriving the OLS [MoM]

$$E[u|x] = 0 \Rightarrow \begin{cases} E[xu] = 0 \\ E[u] = 0 \end{cases} \Leftrightarrow \begin{cases} E[x(y - \beta_0 - \beta_1 x)] = 0 \\ E[y - \beta_0 - \beta_1 x] = 0 \end{cases}$$

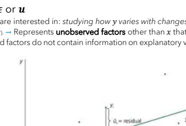
→ Since there are 2 unknown parameters to estimate, we might hope that previous equations can be used to obtain good estimators for the parameters β_0 and β_1
→ Assuming a random sample of n individuals and using the MoM we have:

$$n^{-1}\sum x_i(y - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \text{ and } n^{-1}\sum(y - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

The OLS estimator in the SLRM
Using the basic properties of the summation operator we have:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} \text{ and } \hat{\beta}_1 = \frac{n^{-1}\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n^{-1}\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{Cov(X,Y)}{Var(X)} \rightarrow \text{OLS Estimator}$$

- The **residual** for observation i can be defined as $\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$
  Fit as good as possible a regression line through the data points:



Minimizing the SSR Put otherwise, we choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to make the sum of squared residuals, $SSR(\hat{\beta}_0, \hat{\beta}_1)\sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$
The **foc** for the **minimum** are found by setting the **partial derivatives** equal to **zero**: $\frac{\delta}{\delta\hat{\beta}_0} SSR(\hat{\beta}_0, \hat{\beta}_1) = -2\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$

Normal Equations And these yield so called NE: $n\hat{\beta}_0 + \hat{\beta}_1\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$; $\hat{\beta}_0\sum_{i=1}^n x_i + \hat{\beta}_1\sum_{i=1}^n x_i^2 = \sum_{i=1}^n y_i x_i$ $\quad \frac{\delta}{\delta\hat{\beta}_1} SSR(\hat{\beta}_0, \hat{\beta}_1) = -2\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)x_i$

Goodness of fit: The R^2 The R-square (coefficient of determination):

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SS_{Residuals}}{SS_{Total}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} = \frac{Var(\hat{y})}{Var(y)} = 1 - \frac{Deviance/RSS}{sd^2_{inicial} \times n}$$

- Ratio of the explained variation compared to the total variation;
- It is interpreted as the fraction of the sample variation in y that is explained by x.
- R^2 nearly equal to **0** = a **poor fit** of the *OLS line*: very little of the variation in the y_i is captured by the variation in the $\hat{y}_i$.

› Variance: $Var = s^2 = \frac{\sum_{n=1}^N (y_i - \bar{y})^2}{n}$ › Stand. Deviation: $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$
› Deviance: $DEV^{met} = \sum_{n=1}^N (y_i - \bar{y})^2 = Var \times (n-1)$

Functional Form

Fortunately, it is rather easy to incorporate many **nonlinearities** into simple regression analysis by appropriately defining the dependent and independent variables → *log* transformation is one of the most applied transformations for economic variables.

Table 2: Summary of Functional Forms Involving Logarithms

Model	Dep. Variable	Regressor	Interpretation of β_1	
Level-level	y	x	$\Delta E[y \mid x_i] = \beta_1 \Delta x$	"If x increases one unit, it is expected that the y increases $\hat{\beta}_1$ units."
Log-level	$\log(y)$	x	$\%\Delta E[y \mid x_i] \approx 100\beta_1 \Delta x$	"If x increases one unit, it is expected that the y increases $\hat{\beta}_1 \times 100$ %."
Log-log	$\log(y)$	$\log(x)$	$\%\Delta E[y \mid x_i] \approx \beta_1 \% \Delta x$	"If x increases 1%, it is expected that the y increases $\hat{\beta}_1$ %."
Level-Log	y	$\log(x)$	$\Delta E[y \mid x_i] \approx (\beta_1/100)\%\Delta x$	"If x increases 1%, it is expected that the y increases $\hat{\beta}_1/100$ units."

Remark: If the parameter are estimated (which is always the case) we need to add the term "estimated". For example, ***"The estimated change in E[yi | xi] ..."***.

Multiple Linear Regression Model $\quad Y = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \epsilon$ or u_i

β_0 is still the intercept, $\beta_1, \ldots, \beta_k$ denote slope parameters

Because $\frac{\partial}{\partial x_j} E[y|\mathbf{X}] \approx \beta_j \rightarrow \beta_j$ represents the marginal effect of x_j [Indicates the amount y is expected to change as x_j changes by 1 unit and other variables are kept constant (ceteris paribus)]

Interpreting the OLS $\hat{y}_i = \hat{\beta}_1 \Delta x_{i1} + \ldots + \hat{\beta}_k \Delta x_{ik}$ | In particular, when $x_2, \ldots, x_k$ are held fixed, so that $\Delta x_j = 0$, for $j = 2, \ldots, k$, we have: $\Delta\hat{y}_i = \hat{\beta}_1 \Delta x_{i1}$
The residuals for observation i is: $\hat{u}_i = y_i - \hat{y}_i$

Again, the following properties hold: **1.** $\bar{y} = \bar{\hat{y}}$ **2.** $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1\bar{x}_1 + \ldots + \hat{\beta}_k\bar{x}_k$ **3.** The sample **covariance** between each x_j, for $j = 1, \ldots, k$ and $\hat{u}_i$ **is 0**. Hence, the sample covariance between $\hat{y}_i$ and $\hat{u}_i$.

Assumptions

MLR1: Linear in Parameters The model in the population can be written as $y_i = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_k x_{ik} + u_i = \mathbf{x}_i'\boldsymbol{\beta} + u_i$
MLR2: Random Sampling We have a random sample of n observations, $\{(x_{i1}, x_{i2}, \ldots, x_{ik}), y_i)\}\ i = 1, \ldots, n\}$ following the population model in MLR1.
MLR3: No Perfect Collinearity In the sample there are no exact linear relationships among the independent variables. → **Matrix X has rank $k + 1$.**
MLR4: Zero Conditional Mean Conditional on the entire matrix **X**, each error u_i has zero mean: $E[u_i|\mathbf{X}] = 0$
MLR5: Homoskedasticity The error u has the same variance given any value of the explanatory variables: $Var[u|\mathbf{X}] = \sigma^2 = E[u|X]^2$
In matrix form: $Var(u|\mathbf{X}) = \sigma^2\mathbf{I}_n$ where $\mathbf{I}_n$ denotes the n-dimensional identity matrix.
MLR6: Normality We have: $u_i|\mathbf{X} \sim N(0, \sigma^2)$ [Quando n **grande** uso apenas **TLC** | c.c. convém não ser muito ≠ da N]

Assumptions MLR1-5 → Gauss-Markov Assumptions
Assumptions MLR1-6 → Classical Assumptions or Classical Linear Model Assumptions (CLM Assumptions)

Theorems

1. Unbiasedness of OLS Under assump. MLR1-4 the OLS estimators are unbiased estimators of the population parameters. We have: $E[\hat{\beta}|\mathbf{X}] = \beta$
2. Variances of OLS estimators Under assumptions MLR1-MLR5 the variance-covariance matrix of the OLS estimators is: $Var(\hat{\beta}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$
3. Gauss-Markov Theorem Under assumptions MLR1 - MLR5, the OLS estimators are the **b**est **l**inear **u**nbiased **e**stimators (**BLUE**) of the regression coefficients, i.e $Var(\hat{\beta}_j|\mathbf{X}) \leq Var(\tilde{\beta}_j|\mathbf{X})$ where $\tilde{\beta}_j$ is any unbiased estimator linear in y.
In the matrix sense $Var(\hat{\beta}|\mathbf{X}) \leq Var(\tilde{\beta}|\mathbf{X})$ means that $Var(\tilde{\beta}|\mathbf{X}) - Var(\hat{\beta}|\mathbf{X})$ is a psd matrix.

Inference

Statistical inference in the regression model: • *Hypothesis tests* about **population parameters** • Construction of confidence intervals
Sampling distributions of the OLS estimators: • The OLS estimators are **random variables** | We already know their expected **values** and their **variances**
- However, for **hypothesis tests** we need to know their distribution | In order to derive their distribution we need additional assumptions about distribution of u_i: $N(\mu, \sigma)$
- **Under normality**, **OLS** is **BUE** (For the purposes of statistical inference, the assumption of normality **can be replaced by a large sample size**)

t Test

Under MLR1-6: $\hat{\beta}_j|\mathbf{X} \sim N(\beta_j, Var(\hat{\beta}_j|\mathbf{X}))$ | Therefore, $z = \frac{\hat{\beta}_j - \beta_j}{\sqrt{Var(\hat{\beta}_j|\mathbf{X})}} \sim N(0,1)$ | $t = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{Var}(\hat{\beta}_j|\mathbf{X})}} \sim t_{(n-k-1)} \quad n-(k+1)$

General Setup: Suppose that we have a **hypothesis** about the β_j $\quad H_0: \beta_j = a$ | $H_1: \beta_j \neq a$. where a is a specific value, say $a = 0$

Two possibilities: If $|t_{obs}| > t_{\alpha/2} \Rightarrow$ reject H_0 at the $\alpha \times 100\%$ level | If $|t_{obs}| \leq t_{\alpha/2} \Rightarrow$ do not reject H_0 $\quad$ Where $t_{obs} = \frac{\hat{\beta}_j - a}{se(\hat{\beta}_j)}$ and $t_{\alpha/2}: P(t > t_{\alpha/2}) = \alpha/2$

Under the null hypothesis we have $t = \frac{\hat{\beta}_j - a}{se(\hat{\beta}_j)} \sim t_{(n-k-1)}$
- If we observe $|t_{obs}| > t_{\alpha/2}$ and the H_0 is **true**, then a low-probability event has occurred.
- We take $|t_{obs}| > t_{\alpha/2}$ as an evidence against the null and the decision should be to reject H_0.

Other cases:
- $H_1: \beta_j > a$: if $t_{obs} > t_\alpha$ ⟹ reject the null at the $\alpha \times 100\%$ level; otherwise do not reject.
- $H_1: \beta_j < a$: if $t_{obs} < -t_\alpha$ ⟹ reject the null at the $\alpha \times 100\%$ level; otherwise do not reject.

Calculate *p*-value for $H_0: \beta_j = a$ $\quad H_1: \beta_j \neq a \Rightarrow$ p-value $= 2P(t > |t_{obs}||H_0$ is true) | $H_1: \beta_j > a \Rightarrow$ p-value $= P(t > t_{obs}|H_0$ is true) | $H_1: \beta_j < a \Rightarrow$ p-value $= P(t < t_{obs}|H_0$ is true).
F.e., a $p - value = 0.02$ shows little evidence supporting H_0 → At the 5% level **you should reject the null**.

Correct wording in reporting the outcome of a test involving $H_0: \beta_j = a$ vs $H_1: \beta_j \neq a$
- When the null is rejected we say that $\hat{\beta}_j$ (not β_j) is significantly different from a at $\alpha \times 100\%$ level.
- When the null isn't rejected we say that that $\hat{\beta}_j$ (not β_j) is not significantly different from a at $\alpha \times 100\%$ level.

Correct wording in reporting the outcome of a test involving $H_0: \beta_j = 0$ vs $H_1: \beta_j \neq 0$
- When the null is rejected we say that $\hat{\beta}_j$ (not β_j) is significantly different from **0** at $\alpha \times 100\%$ level, or the regressor (on β_j) is relevant to explain $E[y|\mathbf{X}]$.
- When the null isn't rejected we say that $\hat{\beta}_j$ (not β_j) is not significantly different from **0** at $\alpha \times 100\%$ level, or the regressor (on β_j) is not relevant to explain $E[y|\mathbf{X}]$.

Rejection of the null is not proof that the null is false./ Acceptance of the null is not proof that the null is true. Why?
We prefer to use the language we fail to reject H_0 at the $\alpha \times 100\%$ level rather than H_0 is accepted at the $\alpha \times 100\%$ level.
- In a test of type $H_0: \beta_j = 0$, if $se(\hat{\beta}_j)$ is large ($\hat{\beta}_j$ is an **imprecise estimator**) is more difficult to reject the null. The sample contains little information about the true value of β_j; In a test of type $H_0: \beta_j = 0$ vs $H_1: \beta_j < 0$ (or $H_1: \beta_j > 0$), if $se(\hat{\beta}_j)$, p-values must be divided by two.

Testing a linear **Combination of Parameters** $H_0: \beta_1 = \beta_2 \Leftrightarrow \beta_1 - \beta_2 = 0$ vs $H_1: \beta_1 < \beta_2 \Leftrightarrow \beta_1 - \beta_2 < 0$ | Under the null we have $t = \frac{\hat{\beta}_1 - \hat{\beta}_2}{se(\hat{\beta}_1 - \hat{\beta}_2)}$

Problem: $se(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{\hat{Var}(\hat{\beta}_1) + \hat{Var}(\hat{\beta}_2) - 2 \times \hat{cov}(\hat{\beta}_1, \hat{\beta}_2)}$ (Obter na matriz Cov) $\neq se(\hat{\beta}_1) - se(\hat{\beta}_2)$! | Calculate $\hat{cov}(\hat{\beta}_1, \hat{\beta}_2)$ Just need to pick up element (2,3)

Calculate t_{obs} as $t_{obs} = \frac{0.0667 - .0769}{\sqrt{0.0068^2 + 0.0023^2 - 2 \times 1.927929 \times 10^{-6}}} = \frac{-0.01018}{0.006936} = -1.468$ | The rejection region is: $W = \{t: t_{obs} < -1.645\}$
∴ The p-value is about 0.070, we do not reject the null at the 5% level. There is no evidence against $\beta_1 = \beta_2$ at the 5% level.

F Test $H_0: \beta_{k-q+1} = \beta_{k-q+2} = \ldots = \beta_k = 0$ Which puts q exclusion restrictions on the model.

Testing a linear combination of **Parameters** $\beta_1, \ldots, \beta_k$
- When we impose the restrictions under H_0, we are left with the Restricted Model $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_{k-q} x_{k-q} + u$
  Thus the **restricted model** (**UR**) is a special case of the **unrestricted one** (**R**).
- After estimating both models (**UR** and **R**), what we need to decide is whether the increase in the **SSR** (or in the R^2) in going from the unrestricted model to the restricted model is large enough to warrant rejection of the null. As with all testing, the answer depends on the significance level of the test

$$F = \frac{(SSR_R - SSR_{UR})/q}{SSR_{UR}/(n-k-1)} = \frac{(R^2_{UR} - R^2_R)/q}{(1 - R^2_{UR})/(n-k-1)} \sim F_{(q,n-k-1)}$$

where: $q = df_{H_0} - df_{H_1} = df_R - df_{UR}$ (nº of restrictions under H_0) $n - k - 1 = df_{UR}$ (df of **UR** model)
[**F nunca será 2-sided**]

Let F_{obs} be the observed test statistics. We have: reject H_0 if $F_{obs} > F_\alpha$ (or p-value $< \alpha$) | do not reject H_0 if $F_{obs} \leq F_\alpha$ (or p-value $\geq \alpha$)
Under H_0 we have $F \sim F_{(k,n-k-1)}$ | If $F > F_\alpha$ and the H_0 is **true**, then a low-probability event has occurred. [$p - value \Rightarrow P(F > F_{obs}|H_0$ is true)]

Testing multiple linear **Restrictions**
When $H_0: \beta_1 = \beta_2 = \ldots = \beta_k = 0$ → We are testing of **overall significance of a regression**
The **RM** is $y = \beta_0 + u$ | The F-statistic becomes $F = \frac{SSE/k}{SSR/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)} \sim F_{(k,n-k-1)}$

Heteroskedasticity $Var(u|\mathbf{X}) \neq \sigma^2 = \sigma_i^2$ [$Var(u)$ is different for different values x, then the errors are **heteroskedastic**]

Consequences
1. The *Gauss-Markov Theorem* does not hold (OLS is not BLUE) | **2.** The estimators of the variances $Var(\hat{\beta}_j|\mathbf{X})$ are **biased**
3. Since the OLS standard errors are based directly on these variances, they are no longer valid for constructing **CI** and ***t* statistics**.
4. The usual OLS ***t* statistics** do not have *t* distributions | **5.** **F statistics** are no longer **F distributed** (& **LM statistic** no longer has an asymptotic $\chi^2_{(n)}$ dist.)
In summary, the statistics we used to test hypotheses under the Gauss-Markov assumptions are not valid in the presence of heteroskedasticity

However: OLS estimators are still unbiased. OLS estimators are still consistent

Breusch-Pagan Test [H_0: Var(u | **X**) $= \sigma^2$ erros **homocedásticos** vs H_1: Var(u | **X**) $= \sigma_i^2$ erros heterocedásticos]
1. Estimate the model $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u$ by OLS and obtain the residuals $\hat{u}$;
2. Run the regression $\hat{u}^2 = \alpha_0 + \alpha_1 x_1 + \ldots + \alpha_k x_k + error$ and keep the associated R^2, say $R^2_{\hat{u}}$;
3. Test $H_0: \alpha_1 = \alpha_2 = \ldots = \alpha_k = 0$ against a bilateral alternative, Using the **F-Stat** or the LM, $LM = n \times R^2_{\hat{u}} \overset{a}{\sim} \chi^2_{(k)}$

```
# BP test
library(lmtest)
bptest(reg)
##  studentized Breusch-Pagan test
##
## data:  reg
## BP = 14.092, df = 3, p-value = 0.002782
```

White (Full) Test
Consider, without any lost of generality, $k = 3$: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$
The auxiliary regression is: $\hat{u}^2 = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_1^2 + \alpha_5 x_2^2 + \alpha_6 x_3^2 + \alpha_7 x_1 x_2 + \alpha_8 x_1 x_3 + \alpha_9 x_2 x_3 + \epsilon$
- With only **3 independent** variables the **auxiliary** has 9 independent variables. ($6 \Rightarrow 27$)

```
# Manual regression of squared residuals
summary(lm( resid(reg)^2 ~ lotsize+sqrft+bdrms, data=hprice1))
```

White (Special) Test
Considers the following auxiliary regression

$$\underset{\text{fitted values}}{\hat{u}^2} = \alpha_0 + \alpha_1\hat{y} + \alpha_2\hat{y}^2 + error$$

We can preserve the spirit of the White test while conserving on degrees of freedom by using the OLS fitted values in a test for heteroskedasticity

```
# Manual regression of squared residuals
summary(lm( resid(reg)^2 ~ lotsize+sqrft+bdrms+I(lotsize^2)+
          I(sqrft^2)+I(bdrms^2)+I(bdrms*sqrft)+
          I(bdrms*lotsize)+I(lotsize*sqrft),
# Compute LMobs
length(reg$res)*summary(whitefull)$r.squared
#or
nobs(reg)*summary(whitefull)$r.squared
## [1] 33.73166
```

Robust Estimation
To account for heteroskedasticity in the model, we must resort to a robust variance-covariance matrix to perform statistical inference.

```
# Refined White heteroscedasticity-robust SE:
coeftest(reg, vcov=hccm)
```

robusto, posso usar c/ heteroc.

```
bptest(reg, ~ fitted(reg) + I(fitted(reg)^2) )
##
## studentized Breusch-Pagan test
##
## data:  reg
## BP = 16.268, df = 2, p-value = 0.0002933
```

OLS Asymptotics
Properties of OLS that hold for any sample/sample size Expected values/unbiasedness under MLR.1-4 | Variance formulas under MLR.1-5 | Gauss-Markov Theorem under MLR.1-5 | Exact sampling distributions/tests under MLR.1-6
Properties of OLS that hold in large samples: Consistency under MLR.1-4 | Asymptotic normality/tests under MLR.1-5
Practical Consequences In large samples, the $t - distribution$ is close to the $N(0,1)$ → As a consequence, t-tests are valid without MLR.6
The same is true for confidence intervals and F-tests | **Important**: MLR.1-5 are still necessary, esp. homoskedasticity

Ramsey's RESET Test
Adds **polynomials** in the OLS fitted values to detect general kinds of functional form **misspecification**
1. Estimate the model $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u$ and obtain the fitted values $\hat{y}_i$
2. Estimate augmented model $y_i = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + \gamma_1\hat{y}_i^2 + \gamma_2\hat{y}_i^3 + u_i$
3. Test the null $H_0: \gamma_1 = \gamma_2 = 0$ using an F-test with 2 and $n - k - 3$ **dof** (or an **LM** with **2 dof**)
   Se a eq. inicial for **suficiente** eu irei Fail to Reject

```
data(hprice1, package='wooldridge')
# original linear regression
orig <- lm(price ~ lotsize+sqrft+bdrms, data=hprice1)
# regression for RESET test
RESETreg <- lm(price ~ lotsize+sqrft+bdrms+I(fitted(orig)^2)+
               I(fitted(orig)^3), data=hprice1)
# RESET test
library(lmtest)
resettest(orig)
##
##  RESET test
##
## data:  orig
## RESET = 4.6682, df1 = 2, df2 = 82, p-value = 0.01202
```

Dummys $H_0: \delta_0 = 0$ vs $H_1: \delta_0 \neq 0$

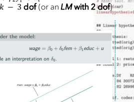
D. Trap Perfect Collinearity [Col linearmente dep.→ **Falha Assup.**]

Discrimination against women is $H_1: \delta_0 < 0$
- Has the dependent variable appearing in *logs*

The coefficient on the dummy has a **percentage interpretation**. $\delta_0 \times 100$ is the percentage difference in a woman's wage in relation to that of a man.

Recall that the *log* approximation may be inaccurate if the percentage difference is big.
- A more accurate approximation is $100(e^{\delta_0} - 1)$
  - We can use several **dummy independent variables** in the same equation.
    F.e., we could add the dummy variable *married*: $sal = \beta_0 + \delta_0 fem + \delta_1 married + \beta_1 educ + u$
    Again, the introduction of these dummies changes the model intercept:

Consider the model: $wage = \beta_0 + \delta_0 fem + \beta_1 educ + u$
Provide an interpretation on δ_0.



	Male	Female
Married	$\beta_0 + \delta_1$	$\beta_0 + \delta_0 + \delta_1$
Single	β_0	$\beta_0 + \delta_0$

4. Panel Data Pooled OLS. Fixed and Random Effects

PD have a ***cross-sectional*** and a ***time series*** dimension, where **CS units are followed over time.** | Can be used to account for time-invariant unobservables.
For a linear regression model with panel data, we have: $y_{it} = \beta_0 + \beta_1 x_{it} + u_{it}$ where y_{it} and x_{it} are observed for $i = 1, \ldots, N$ individuals (countries, ...) over $t = 1, \ldots, T$ time periods (days, months, quarters, years, ...)

Why not consider the univariate cross-sectional linear regression model? The main challenge in identifying the effect of x on y from a **univariate cross-sectional model is that individuals** in the sample **not only differ in terms of x**, but also differ along other dimensions.
- Solve the **Omitted Variables Problem**. Let y and $x = (x_1, x_2, \ldots, x_k)$ be observable **r.v.**, and let c be an **unobservable r.v.** or **heterogeneity**.
  As is often the case, we are interested in the partial effects of the observable explanatory variable x_j in the population regression function: $E[y|x_1, x_2, \ldots, x_k, c]$
  In other words, we would like to hold c constant when obtaining partial effects of the observable explanatory variables.
- Assuming a linear model, with c entering additively along with the x_j, we have: $E[y|\mathbf{x}, c] = \beta_0 + \mathbf{x}\beta + c$
  Situação 1: If c is uncorrelated with each x_j, then c is just another unobserved factor affecting y that is not systematically related to the observable explanatory variables whose effects are of interest.
  Situação 2: if $Cov(x_j, c) \neq 0$ for some j, putting c into the error term can cause some problems.

```
#Pooled ols
library(plm)
pool <- plm(lwage ~ educ + exper
   + married + union, data=wagepan, model='pooling')
summary(pool)
```

Estimators

Pooled OLS Estimation simply applies the OLS for a sample of $N \times T$ observations, ignoring the panel structure of the data.
- Under certain assumptions, the **POLS estimator** can be used to obtain a consistent estimator of β in the model below:
  $y_{it} = \mathbf{x}_{it}\beta + v_{it}, t = 1, \ldots, T, i = 1, \ldots, N$ where $v_{it} = c_i + u_{it}$ are the **composite errors**
  - For each t, v_{it} is the **sum of the unobserved effect and an idiosyncratic error** (u_{it}).

Pooled OLS estimation *is appropriate to use when the regressors are not correlated with c_i*

Pooled OLS Assumptions
POLS.1: Linearity The data sequence $\{y_{it}, x_{1it}, x_{2it}, \ldots, x_{kit}\}$ follows the linear model: $y_{it} = \mathbf{x}_{it}\beta + v_{it}, t = 1, \ldots, T, i = 1, \ldots, N$ where $\{v_{it}\}$ is the sequence of composite errors.
POLS.2: Random Sampling The individuals are select by random sampling. This implies that $\{y_i, x_i\}$ are i.i.d., i.e., the observations are independent across individuals but not necessarily across time.
POLS.3: Contemporaneous exogeneity $E[v_{it}|\mathbf{x}_{it}] = \mathbf{0}$, which implies that $E[\mathbf{x}_{it}c_i] = 0$ and $E[\mathbf{x}_{it}u_{it}] \Rightarrow E[\mathbf{x}_{it}v_{it}] = 0$
POLS.4: No Perfect Collinearity In the sample, no independent variable is constant nor a perfect linear combination of the others.
Consistency Knowing that we have a sample of N independent cross sections observed during T periods of time and that c_i is independent of u_{it}, under POLS1-4, the pooled OLS is **consistent**, however if c is correlated with any element of $\mathbf{x}_t$, then **POLS** is biased and **inconsistent**.
POLS.5: Homoskedasticity and No Serial Correlation $Var(u_i|\mathbf{x}_i, c_i) = \sigma_u^2\mathbf{I}_T$
- However, in this setting, the composite errors v_{it} will be serially correlated due to the presence of c_i in each time period.
- Therefore, inference using pooled OLS requires the **robust variance matrix estimator**.

Random Effects Estimation As the **POLS**, **RE** puts c_i into the error term.
The rationale behind random effects model is that the variation across individuals is assumed to be random and uncorrelated with the predictor or independent variables included in the model.

Random Effects estimation *is appropriate when c_i is random and influences the dependent variable but is not correlated with the independent variables.*

Random Effects Properties
RE.1: Linearity The data sequence $\{y_{it}, x_{1it}, x_{2it}, \ldots, x_{kit}\}$ follows the linear model: $y_{it} = \mathbf{x}_{it}\beta + v_{it}, t = 1, \ldots, T, i = 1, \ldots$ where $\{v_{it}\}$ is the sequence of composite errors.
RE.2: Random Sampling The individuals are select by random sampling. **[=POLS]**
- Random effects estimation imposes more assumptions than those needed for **POLS**: **strict exogeneity** in addition to orthogonality between c_i and $\mathbf{x}_{it}$.

RE.3: Strict exogeneity (a) $E[u_{it}|\mathbf{x}_i, c_i] = 0, t = 1, \ldots, T$ [Assume $Corr(ability, educ) = 0$] $\quad$ (b) $E[c_i|\mathbf{x}_i] = E[c_i] = 0$ where $\mathbf{x}_i = (\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots, \mathbf{x}_{iT})$
- The random effects approach exploits the serial correlation in the composite error, $v_{it} = c_i + u_{it}$, in a generalized least squares (GLS) framework.
  For consistency of GLS, we need the usual rank condition for GLS: **RE.4: No Perfect Collinearity** $rank(E[\mathbf{X}_i'\Omega^{-1}\mathbf{X}_i]) = K$

RE.5: Homoskedasticity (a) $E[\mathbf{u}_i\mathbf{u}_i'|\mathbf{x}_i, c_i] = \sigma_u^2\mathbf{I}_T$ (b) $E[c_i^2|\mathbf{x}_i] = \sigma_c^2$
Consistency and Efficiency
Let $\hat{\beta}_{RE}$ be the random effects estimator: Under assumptions **RE.1-4** ⇒ $\hat{\beta}_{RE}$ is consistent. Under assumptions **RE.1-5** ⇒ $\hat{\beta}_{RE}$ is asymptotically efficient.

```
library(plm)
library(wooldridge)
re <- plm(lwage ~ educ + exper
   + married + union, model='random', data=wagepan)
summary(re)
```

Fixed Effects Estimation When the c_i may be correlated with the explanatory variables, we use the **Fixed Effects estimation**.
FE.1: Linearity The data sequence $\{y_{it}, x_{1it}, x_{2it}, \ldots, x_{kit}\}$ follows the linear model: $y_{it} = \mathbf{x}_{it}\beta + c_i + u_{it}, t = 1, \ldots, T, i = 1, \ldots, N$
FE.2: Random Sampling The individuals are select by random sampling. This implies that $\{y_i, x_i\}$ are i.i.d., i.e., the observations are independent across individuals but not necessarily across time.
But relax assumption: FE.3: Strict Exogeneity $E[u_{it}|\mathbf{x}_i, c_i] = 0, t = 1, 2, \ldots, T$
- In **FE** analysis, $E[c_i|\mathbf{x}_i]$ is allowed to be any function of $\mathbf{x}_i$. Essentially, which allows to consistently estimate partial effects in the presence of time-constant omitted variables that can be arbitrarily related to the observable $\mathbf{x}_{it}$. Therefore, **FE analysis is more** robust **than RE analysis**.

The idea for estimating β under the Assumption FE.3 is to transform the equations to eliminate the unobserved effect c_i. This transformation is denoted the **fixed effects transformation** or **within transformation**.

Considering the mean values **within** each individual: $\bar{y}_i = T^{-1}\sum_{t=1}^T y_{it}$ $\bar{\mathbf{x}}_i = T^{-1}\sum_{t=1}^T \mathbf{x}_{it}$ $\bar{u}_i = T^{-1}\sum_{t=1}^T u_{it}$
Subtracting these terms for each t gives the FE transformed equation: $y_{it} - \bar{y}_i = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\beta + u_{it} - \bar{u}_i$ or $\ddot{y}_{it} = \ddot{\mathbf{x}}_{it}\beta + \ddot{u}_{it}, t = 1, 2, \ldots, T$
Downside: cannot be used to investigate time-invariant causes of y

Since the assumption $E[\ddot{\mathbf{x}}_{it}'\ddot{u}_{it}] = 0$ holds under Assumption FE.3, we can apply **POLS**. The **FE estimator**, denoted by $\hat{\beta}_{FE}$ is the **POLS** estimator from the regression $\ddot{y}_{it}$ on $\ddot{\mathbf{x}}_{it}$.
FE.4: No Perfect Collinearity $rank(\sum_{t=1}^T E[\ddot{\mathbf{x}}_{it}'\ddot{\mathbf{x}}_{it}]) = k$
FE.5: Homoskedasticity $E[\mathbf{u}_i\mathbf{u}_i'|\mathbf{x}_i, c_i] = \sigma_u^2\mathbf{I}_T$ It can be shown that the errors ($\ddot{\mathbf{u}}_{it}$) are **negatively** serially correlated, however as T gets large, the correlation tends to 0. | Consistency Under **FE.1-4**, the **fixed effects estimator is consistent**.

Pooled OLS
- c_i is not correlated with x_{it}
- Ignore panel data structure and estimate through O
- Under POLS.1 to POLS.4, is consistent

Fixed effects
- c_i is correlated with x_{it}
- Demean each variable in the regression and estimate through OLS
- Under FE.1 to FE.4, is consistent

Random effects
- c_i is not correlated with x_{it}
- Consider correlation structure and estimate through GLS
- Under RE.1 to RE.5, is consistent and efficient

Hausman Test $H_0: E[x_{it}c_i] = 0$ (both **FE** and **RE** are consistent **estimators**, **RE** is the **most** efficient **estimator**) **vs** $H_1: E[x_{it}c_i] \neq 0$ (only **FE** is consistent)
- The test applies only to variables that vary on time.
- If the assumption **RE.4** fails, **one must use the** Robust Hausman test.

5. Causal Inference

Is about *counterfactual prediction*, predict what would have happened to the same units/subjects had they were exposed to a different (counterfactual) condition.

Definition 1. Individual treatment effect (δ_i) is equal to $Y_i^1 - Y_i^0$ [Para o indivíduo i | **Não é** algo estimado]

Definition 2. Switching equation

An individual's realized health outcome, Y_i, is determined by treatment assignment, D_i which selects one of the potential outcomes:

$$Y_i = D_iY_i^1 + (1-D_i)Y_i^0 \Rightarrow Y_i = \begin{cases} Y_i^1 \text{ if } D_i = 1 \\ Y_i^0 \text{ if } D_i = 0 \end{cases}$$

Not the same as treatment assignment mechanism. **Treatment** assignment describes how D was assigned, not whether it was assigned.

Definition 3. Fundamental problem of causal inference If you need both potential outcomes to know causality with certainty, then since it is impossible to observe both Y_i^1 and Y_i^0 for the same individual, δ_i, is *unknowable*.

Definition 4. Average treatment effect (ATE) (não obs.) is the population average of all i individual treatment effects $E[\delta_i] = E[Y_i^1 - Y_i^0] = E[Y_i^1] - E[Y_i^0]$
- Aggregate parameters based on individual treatment effects are *summaries* of individual treatment effects
- Cannot be calculated because Y_i^1 and Y_i^0 do not exist for the same unit i due to switching equation

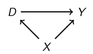
Definition 5. Average Treatment Effect on the Treated (ATT)

The average treatment effect on the treatment group is equal to the average treatment effect conditional on being a treatment group member: $E[\delta|D=1] = E[Y^1 - Y^0|D=1] = E[Y^1|D=1] - E[Y^0|D=1]$
- Cannot be calculated because Y_i^1 and Y_i^0 do not exist for the same unit i due to switching equation.

Definition 6. Average Treatment Effect on the Untreated (ATU)

The average treatment effect on the untreated group is equal to the average treatment effect conditional on being untreated: $E[\delta|D=0] = E[Y^1 - Y^0|D=0] = E[Y^1|D=0] - E[Y^0|D=0]$
- Cannot be calculated because Y_i^1 and Y_i^0 do not exist for the same unit i due to switching equation

Directed Acyclic Graphs - DAGs
- Treatment (e.g. intervention, exposure): D *ter mestrado*
- Outcome (e.g. disease status): Y *salário*
- Observed **covariates or** confounders: X *salários dos pais (que afeta D e Y)*
- Unobserved **covariates or** confounders: U

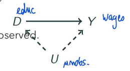


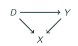
- **3 random variables:** X, D and Y.
- There is a **direct path from D to Y**, which represents a **causal effect**.
- But there is also a 2nd path **from D to Y** called the *backdoor path*
- The **backdoor path** is given by $D \leftarrow X \rightarrow Y$. The **BP** is not causal, we therefore call X a confounder.

Confounding occurs when the **treatment** and the **outcomes** have a **common cause** or parent which creates spurious correlation between D and Y.
- The correlation between D and Y **no longer reflects the causal effect** of D on Y because of selection bias due to a confounder.
- We need then to control for X to **close** the backdoor path.

We can **"block"** any particular backdoor path by conditioning on variable X so long as it is not a **collider** (later)
- → Once we condition on X, the correlation between D and Y estimates the causal effect of D on Y
- → Conditioning means calculating $E[Y|D=1,X] - E[Y|D=0,X]$ for each value of X

DAG with Unobserved Variables Now we have a confounder that is unobserved.
- Since U is unobserved, that backdoor pathway is **open**.





X é collider, *está a ser causado* D, Y

não podemos fixar em U, vamos usar IV

- $D \rightarrow Y$ (causal effect of D on Y)
- $D \rightarrow X \leftarrow Y$ (backdoor path 1)

Collider = ant. There are 2 ways to get from D to Y | **But** X **has 2 arrows pointing to it, not away from it.**
- When 2 variables cause a third variable along some path, we call that third variable a "collider".

Colliders are special because when **they appear along a** ***backdoor path***, that backdoor path is closed simply because of their presence.

Why is this important? DAGs allow us to map the relationship between variables.
- Identify colliders and confounding factors will determine which variables we should (and should not) condition, i.e., control for in the analysis. *To identify a causal relationship, we want to block (**close**) all the backdoor paths.*

Blocked Backdoor Paths A backdoor path is blocked if and only if:
- It contains a **noncollider** that has been conditioned on **OR** a **collider** that has not been conditioned on

1. Conditioning on a noncollider blocks a path: [X] ⟶ Z ⟶ Y
2. Conditioning on a collider opens a path (i.e., creates spurious correlations): Z ⟶ [X] ⟵ Y
3. *Not* conditioning on a collider blocks a path: Z ⟶ X ⟵ Y

Backdoor Criterion

Conditioning on X satisfies the backdoor criterion with respect to (D,Y) directed path if:

1. All backdoor paths are **blocked** by X **2.** No element of X is a **collider**

In words: If X satisfies the backdoor criterion with respect to (D, Y), then controlling for or matching on X identifies the causal effect of D on Y. And again note that a path which has a conditioned-on-collider can still be closed, but only with a noncollider-conditioned-on

What control strategy meets the backdoor criterion?

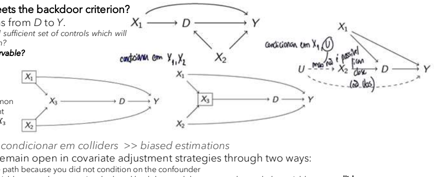
List all backdoor paths from D to Y.

What are the necessary and sufficient set of controls which will satisfy the backdoor criterion?

What if you have an unobservable?

Multiple strategies

Conditioning on the common causes, X_1 and X_2, is sufficient ....but so is conditioning on X_3



Collider Bias *overfit >> condicionar em colliders >> biased estimations*

Backdoor paths can remain open in covariate adjustment strategies through two ways:

a. You did not close the path because you did not condition on the confounder

b. Your conditioning variable opened up a previously closed backdoor path because on that path the variable was a **collider**

Colliders are *"bad controls"*; they are often outcomes, but not always and not all outcomes create this problem

This is the risk of blindly controlling for variables (*"kitchen sink regressions"*)

Instrumental Variables Z, that:

I. influences treatment assignment; D

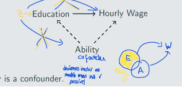
II. is independent of unmeasured confounders; U

III. has *no direct* effect on the outcome Y, except through its effect on the treatment. D

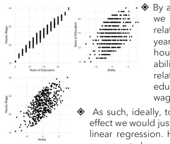
When should we use instrumental variables? **When we have** ***unobserved*** **confounders**; U

Why? Allows to estimate the variation in the treatment that is free of the unmeasured confounders;
- This confounder-free variation in the treatment, except to estimate the causal effect of the treatment.

Suppose you want to identify the causal relationship of years of education on wage. One possible drawback could be due to the presence of an omitted unobservable variable, *ability*, that influences simultaneously education and wage.



Here, *ability* is a confounder.



- By analyzing these plots, we can see a positive relationship between years of education and hourly wage. But also, ability is positively related with years of education and hourly wage.
- As such, ideally, to uncover this causal effect we would just include ability in the linear regression. However, in practice, we cannot observe ability...

Two Stage Least Squares (2SLS)

The fitted value, *educ*, can be seen as the estimated version of *educ* that is *uncorrelated* with u. Therefore, 2SLS first "purges" *educ* of its correlation with u before doing the OLS regression.

Specification Tests

Weak instruments The null hypothesis is H_0: "All instruments are weak". Thus, *rejecting* the null hypothesis indicates the instruments used are ***strong***, i.e., it is highly correlated with the endogenous regressor it concerns. $corr(Z,X)$ *baixa*

Hausman test for endogeneity The null hypothesis is H_0: $Cov(x,u) = 0$. Thus, *rejecting* the null hypothesis indicates the existence of *endogeneity* and the need for instrumental variables. *se ñ ex, ñ preciso IV*

Sargan test This test for the validity of instruments (whether the *instruments* are *correlated* with the error term) can *only* be performed for the ***extra*** **instruments**, those that are in excess of the number of endogenous variables. *queremos que não esteja correlated | quando tal não como não é possivel fazer o teste*
- The null hypothesis is H_0: $Cov(z,u) = 0$. Thus, *rejecting* the null hypothesis indicates that at least one of the extra instruments is not valid. *temos de ter pelo menos tantos IV quantos regressores*

Difference-in-Differences (DiD) is a very old, relatively straightforward, intuitive research design
- A group of units are assigned some ***treatment*** and then compared to a group of units that ***weren't*** | Mostly associated with "big shocks" happening in space over time

Minimum wage on employment - Card and Kruger (1994)

Goal: Evaluate the effect of minimum wage on employment.

Units: fast-food restaurants in New Jersey (NJ) and adjacent eastern Pennsylvania (PA).

Intervention: Raise of the state minimum wage; NJ raised the minimum wage on April 1, 1992, but PA did not. *"treatment" reg.*

Approach: Compare the employment (average per store), observed in both areas, before and after the change.

Outcome: After the increase in minimum wage, NJ's employment increased.

$$y_{it} = \beta_0 + \beta_1 D_i + \beta_2 Post_t + \beta_3(D_i \times Post_t) + u_{it}$$

where $D = 1$ if i is in the treatment group and $Post = 1$ if t is after the treatment. $E[y \mid D=1]$

- We only have two groups (**treated/untreated**) and two periods (**before/after the treatment**)

Equation implies that we have one regression for the **control** group: $D=0$ $y_{it} = \beta_0 + \beta_2 Post_t + u_{it}$

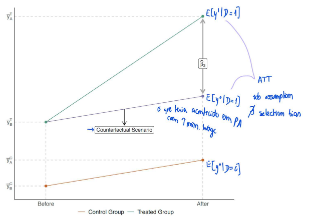
and another one for the **treated** group $D=1$ $y_{it} = (\beta_0 + \beta_1) + (\beta_2 + \beta_3)Post_t + u_{it}$

Since $Post_t$ is a dummy variable, estimating equation through **OLS** would yield:

$$\overline{Y}_{C,B} = \hat\beta_0 \qquad \overline{Y}_{T,B} = \hat\beta_0 + \hat\beta_1 \qquad \overline{Y}_{C,A} = \hat\beta_0 + \hat\beta_2 \qquad \overline{Y}_{T,A} = \hat\beta_0 + \hat\beta_1 + \hat\beta_2 + \hat\beta_3$$

The regression framework allows to add regressors as controls.

In that case, the interpretation is no longer as sample average, but as partial effects after controlling for other factors



The purple line represents the counterfactual **scenario,** i.e., how we would expect the treated group to behave in the absence of treatment.

This is based on a key assumption: the **Parallel Trends Assumption**.

If the *PTA* holds, then the differences in means between the control and treated group, *before* the intervention are **zero**.

Exercises

1. DD (a) If a variable X follows **a Binomial Distribution with only one trial**, then it is equivalent of following a Bernoulli Distribution. - **True** because the $X \sim Ber(p) = Bin(n=1, p)$

(b) A Poisson distribution is appropriate to describe binary outcomes.

False because $Po(\lambda)$ is used to describe the number of events in a fixed interval of time or space.

(c) The probability density function describes the probability function of a discrete variable whereas a probability mass function describe the probability mass function of a continuous random variable.

False because the probability mass function describes the probability function of a discrete variable whereas a probability density function describe the probability mass function of a continuous random variable.

(d) A variable is discrete if the sample space is countable. **False** because a d.v. is a variable that can only take on a certain number of values.

1. DD Let X be a random variable. In R we performed the following instruction:

```
dbinom(4, 10, 0.6)
[1] 0.1115
```

- The probability of getting **4 successes** - x - in **10 trials iid** (independent and identically distributed) - n - with a probability of success of $p = 0.6$ is 0.1115.
  - $P(X \le 5) = 0.3669$
  - $P(X < 5) = P(X \le 4) = 0.1662$
  - $P(X > 4) = 1 - P(X \le 4) = 1 - 0.1662 = 0.8338$
  - $P(X = 5) = P(X \le 5) - P(X < 5) = 0.3669 - 0.1662 = 0.2007$

```
pbinom(4, 10, 0.6)    # = P(X <= 4)
[1] 0.1662
pbinom(5, 10, 0.6)    # = P(X <= 5)
[1] 0.3669
```

1. CD (a) Let $X_1, X_2, \ldots, X_n$ be n independent random variables having the standard normal distribution, then let $Y = \sum_{i=1}^n X_i^2$, then Y follows a chisquared distribution with n degrees of freedom.

True because if $X_1, X_2, \ldots, X_n$ are independent and identically distributed (***iid***) random variables with a standard normal distribution, then $Y = \sum_{i=1}^n X_i^2$ follows a chisquared distribution with n degrees of freedom. I.e, the sum of squares of n independent standard normal random variables follows a chi-squared distribution with n degrees of freedom.

(b) Let $X_1 \sim \chi^2(v_1)$ and $X_2 \sim \chi^2(v_2)$, then $Y = \frac{X_1/v_1}{X_2/v_2}$ follows a t-distribution with $v_1 + v_2$ degrees of freedom.

False The correct distribution for Y is the $F-distribution$ with v_1 and v_2 degrees of freedom. The $t-distribution$ is related to the ratio of a standard normal random variable and the square root of a chi-squared random variable divided by its degrees of freedom - $t = \frac{Z}{\sqrt{X/v}}$

(c) A standard normal distribution has $\mu = 0$ and $\sigma = 1$.

True because a standard normal distribution has a mean of 0 and a standard deviation of 1.

(d) The t-distribution is symmetric and bell-shaped, like the normal distribution, but has heavier tails, meaning that it is more prone to producing values that fall closer to its mean.

False because the t-distribution is symmetric and bell-shaped, like the normal distribution, but has heavier tails, **not meaning** that it is more prone to producing values that fall ~~closer~~ to its mean.

1. CD Consider $X \sim F(4,300)$. `pf(x,4,300)` $P(X > 3) = P(X \ge 3) = 1 - P(X \le 3) = 0.0189$

$P(X < 3) = 0.981$ $P(3 \le X \le 6) = P(X \le 6) - P(X < 3) = 0.0187$

1. CD `pnorm(0.975,1,2)`= $X \sim N(\mu = 1, \sigma^2 = 4) \rightarrow$ mean of $\mu = 1$ and a standard deviation of $\sigma = 2$, so variance is $\sigma^2 = 4$ | Returns the **quantile** value corresponding to the cumulative probability of **0.975** **[qnorm** - gives the 97.5th percentile]

1. CD What is the result of `qnorm(pnorm(2))`

The result is 2 because the **`pnorm(2)`** is the cumulative density function value for the standard normal distribution in $x = 2$, and the **`qnorm()`** is the quantile for the standard normal distribution in $x = 0.9772$.

`pnorm(2)`: This function calculates the cumulative distribution function (**CDF**) of the standard normal distribution (mean = 0, standard deviation = 1) for the value 2.
- In other words, it computes the probability that a value from a standard normal distribution is less than or equal to 2. - $[P(X \le 2)$ where $X \sim N(0,1)]$
- The result is approximately 0.9772, meaning about 97.72% of the distribution falls below 2.

`qnorm(0.9772)`: - This function computes the **quantile** (or **inverse CDF**) of the standard normal distribution for a probability of 0.9772.
- It essentially asks, ***"What value of X from the standard normal distribution has a cumulative probability of 0.9772?"*** - The answer is 2, because $P(X \le 2) = 0.9772$ for a $N(0,1)$.

2. MoM Consider a **R.V.** X whose distribution depends on the parameters α and θ, for which we have $E(X) = \alpha\theta$ and $Var(X) = \alpha\theta^2$. Knowing that in a sample of 320 observations we have $\sum_{i=1}^{320} xi = 22.2$ and $\sum_{i=1}^{320} x^2 i = 535.8$, present a plausible estimate for the unknown parameters. Hint: $Var(X) = E(X^2) - (E(X))^2$

The MoM estimates the parameters of a distribution by setting the sample moments equal to the population moments.

The first moment is given by: $E(X) = \frac{1}{n}\sum_{i=1}^n x_i = \alpha\theta$ | The second moment is given by: $E(X^2) = \frac{1}{n}\sum_{i=1}^n x_i^2 = \alpha\theta^2 + \alpha\theta^2 = 2\alpha\theta^2$

The variance is given by: $Var(X) = E(X^2) - (E(X))^2 \Leftrightarrow \alpha\theta^2 = E[X^2] - (\bar{x})^2 \Leftrightarrow E[X^2] = \alpha\theta^2 + (\alpha\theta)^2$

$\begin{cases} E[X] = \frac{1}{n}\sum_{i=1}^n x_i \\ Var[X] = \frac{1}{n}\sum_{i=1}^n x_i^2 - \left(\frac{1}{n}\sum_{i=1}^n x_i\right)^2 \end{cases} \Leftrightarrow \begin{cases} \alpha\theta = \bar{x} \\ (\alpha\theta^2) + (\alpha\theta)^2 = \frac{\sum x_i^2}{n} \end{cases}$ | Considering $(\alpha\theta)^2 = \bar{x}^2$ and $\frac{\bar{x}}{\theta}\theta^2 = \bar{x}\theta$ we have: $\begin{cases} \alpha = \frac{\bar{x}}{\theta} \\ \bar{x}\theta + \bar{x}^2 = \frac{\sum x_i^2}{n} \end{cases} \Leftrightarrow \begin{cases} \alpha = \frac{\bar{x}}{\theta} \\ \theta + \bar{x} = \frac{\frac{\sum x_i^2}{n}}{\frac{\sum x_i}{n}} \end{cases} \Leftrightarrow \theta = \frac{\sum x_i^2}{\sum x_i} - \bar{x} \Leftrightarrow \theta = \frac{\sum x_i^2}{\sum x_i} - \frac{\sum x_i}{n}$

As $\sum_{i=1}^{320} xi = 22.2$, $\sum_{i=1}^{320} x^2 i = 535.8$ and $n = 320$ we have: $\begin{cases} \alpha = \frac{22.2/320}{\theta} \\ \theta = \frac{535.8}{22.2} - \frac{22.2}{320} \end{cases}$ | So, $\theta = 24.06576$ and $\alpha = 0.002882726$.

2. (a) The primary goal of **point estimation** is to find a range of possible values for a population parameter.

False ... Is to find a single value that is the best estimate of the pop. parameter, not a range of possible values.

(b) Both the **MoM** and **MLE** can be used to estimate parameters for **discrete** and **continuous** prob. distri. **TRUE**

(c) For estimating the parameter of a **Poisson** distribution using the method of moments, we need to solve two equations. **False** We only need to solve one equation → the first moment equation $E(X) = \alpha\theta$.

(d) MLE aims to find parameter values that maximize the likelihood function, which measures the probability of observing the given data for a given set of parameter values. **True**

(e) The parameter space, ω, defines the range of possible values for the observed values of a random sample.

False ω defines the range of possible values for the parameters of the distribution, not the observed values of a random sample.

2. HT (a/c) $\mu = 3$ | $\sigma < 3$ Statistical hypothesis → This is a statement about a population parameter

(b/d) $\bar{x} = 4$ | $\bar{X} < 3$ Not a statistical hypothesis → This is a statement about a sample statistic.

2. HT A factory produces a certain synthetic material whose breaking strength follows a random variable with a mean of $250\ kg$. The production process used is expensive, and in an attempt to reduce costs while ensuring the same average level of breaking strength, the factory administration decided to collect a sample of 25 units manufactured by the new process and proceed to test $H_0: \mu \ge 250$ against $H_1: \mu < 250$ with a significance level of 0.025.

(a) Given the proposed test, do you think the company is more inclined to adopt the new process or maintain the old one?

The company is more inclined to **adopt the new process**, as the null hypothesis is that the mean strength is greater than or equal to $250\ kg$.

$H_0: \mu \ge 250$ → states that the new process is at least as good as the old one. | $H_1: \mu < 250$ → suggests that the new process is worse.

By conducting a one-tailed test with a significance level of 0.025, the company is demonstrating a willingness (disposição) **to reduce the risk of incorrectly rejecting a good new process** (**Type I error**) even at the expense of potentially missing a significant improvement from a new manufacturing process.
- In other words, $\alpha = 0.025$ make it's harder to reject the null hypothesis, which means that is harder to reject the new process.
- So the company is more inclined to adopt the new process.

Additional Note: **As the sample size increase, it's easier to reject the null hypothesis.**

(b) Assuming that in the sample it was observed a mean strength of the material of $245\ kg$ and a corrected standard deviation of $6\ kg$, calculate the test value and conclude whether or not to adopt the new manufacturing process.

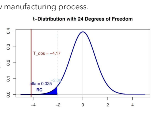
The Test Statistic for the sample mean is given by: $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ where $\bar{x} = 245$, $\mu = 250$, $s = 6$, and $n = 25$.

P-value Calculation: The p-value is the probability of observing a sample mean as extreme or more extreme than the one obtained, assuming the null hypothesis is true.
- In this case, the p-value is calculated using the t-distribution with 24 degrees of freedom $(n-1)$.

$$p - value = P(X < -4.17)$$

Assuming $X \sim t_{24}$, we can calculate the p-value using the cumulative distribution function (CDF) of the *t-distribution*.

```
# P-value calculation
(p_value <- pt(t, df = 24))
## [1] 0.0001727612
```



Conclusion: Using a $t-distribution$ table with 24 degrees of freedom $(n-1)$, we find that the critical value for a one-tailed test at a significance level of 0.025 is approximately −2.49.

Since the calculated t-value (−4.17) is less than the critical value (−2.49), **we reject the null hypothesis**. This means there is sufficient evidence to conclude that the average breaking strength of the material produced by the new process is less than $250\ kg$. Therefore, the company **should not adopt the new manufacturing process**.

Using $p - value$ we can also reject the null hypothesis, since $p - value = 0.0001$ is less than the significance level of 0.025.

2. HT Consider a study to determine whether a sales tax on soda will reduce consumption of soda in the US below the current per-capita level of about 50 gallons of soda per year. The hypotheses for the test are $H_0: \mu = 50$ vs $H_1: \mu < 50$, where μ represents the average annual consumption of soda in communities where the sales tax is implemented.
- In this case, since we **don't know the standard deviation of the population**, the appropriate test statistic is the t-statistic, which follows a **t-distribution with** $n-1$ **degrees of freedom**.

(a) Suppose sample results give a p-value of 0.02. Interpret this p-value in terms of random chance and in the context of taxes and soda consumption.

A $p - value = 0.02$ means that there is a 2% chance of observing a sample mean as extreme or more extreme than the one obtained ($245\ kg$), assuming the null hypothesis ($\mu = 50$) is true.

Very unlikely to happen by random chance → High evidence against the null hypothesis → High $p - value$

In other words, if the sales tax had no effect on soda consumption, there's only a 2 chance of getting a sample that suggests a reduction as strong as the one observed. This provides **strong evidence** against the null hypothesis, suggesting that the sales tax might be effective in reducing soda consumption.

(b) Now suppose sample results give a p-value of 0.41. Interpret this p-value in terms of random chance and in the context of taxes and soda consumption.

A $p - value = 0.41$ means that there is a 41% chance of observing a sample mean as extreme or more extreme than the one obtained, assuming the null hypothesis is true. This is a relatively high probability, indicating that the observed reduction in soda consumption could be due to random chance. Therefore, there is **not enough evidence** to conclude that the sales tax is effective in reducing soda consumption.

(c) Which p-value, 0.02 or 0.41, gives stronger evidence that a sales tax will reduce soda consumption?

The $p - value = 0.02$ **provides stronger evidence** that a sales tax will reduce soda consumption. A lower $p - value$ indicates a lower probability of observing the sample results under the null hypothesis (H_0), making it more likely that the observed effect is due to the treatment (in this case, the sales tax).

2. HT (a) In a test with significance level $\alpha = 0.05$, a p-value of 0.078 leads to not rejecting H_0.

True $p - value >$ $significance\ level$, indicating that there is not enough evidence to reject the null hypothesis (H_0).

(b) When in a hypothesis test a p-value of 0.05 is obtained, one is more confident in rejecting H_0 than when the p-value is 0.01. **False** A p-value of 0.01 is lower than a p-value of 0.05, indicating stronger evidence against the null hypothesis (H_0). Therefore, one is more confident in rejecting H_0 when the p-value is 0.01 than when it is 0.05. In other words, a smaller p-value (e.g., 0.01) indicates stronger evidence against H_0, making one more confident in rejecting it.

(c) If in the test $H_0: \mu = \mu 0$ against $H_1: \mu < \mu 0$, referring to a population with known variance and mean $\mu 0$, the p-value is 0.05, then $P(\bar{X} > \bar{\mu} = \mu_0)$ is 0.95. **True** $p - value = 0.05$, indicating that there is a 5% chance of observing a sample mean as extreme or more extreme than the one obtained, assuming the null hypothesis is **true**.

Since we know population variance and mean, we can calculate the **z-value** and compare it with the **critical value** to reject or not the null hypothesis. $P(Z \le Z_{obs} \mid \mu = \mu_0) = 0.05 \Leftrightarrow P(Z > Z_{obs} \mid \mu = \mu_0) = 0.95 \Leftrightarrow P(\bar{X} > \mu_0) = 0.95$ Assuming that H_0 is **true** and $P(Z > Z_{obs}) = 1 - P(Z \le Z_{obs}) = 0.05$, we have $P(Z \le Z_{obs}) = 0.95$.

(d) In a hypothesis test, the p-value **does not** depend on the test's significance level.

True The p-value is calculated based on the sample data and the null hypothesis, and it does not depend on the chosen significance level.

2. HT (a) A narrower confidence interval indicates greater precision in estimating the population parameter.

True As the range of possible values is smaller.

(b) A $Type\ II\ error$ occurs when you incorrectly reject the null hypothesis when it is true.

False A $Type\ II\ error$ occurs when you fail to reject the null hypothesis when it is actually false. Rejecting a true null hypothesis is a $Type\ I\ error$.

(c) The p-value is the probability of making a $Type\ I\ error$ when the alternative hypothesis is true.

False The $p - value$ is the probability of observing a test statistic as extreme as (or more extreme than) the one calculated, assuming the null hypothesis is true. It does not relate to the probability of Type I error given the alternative hypothesis.

(d) Increasing the sample size generally leads to wider confidence intervals.

False Increasing the sample size generally leads to narrower confidence intervals, as the standard error of the estimate decreases with larger sample sizes.

(e) The power of a statistical test is the probability of correctly rejecting the null hypothesis when it is true.

False The power of a test is the probability of correctly rejecting the null hypothesis when it is false (i.e., detecting an effect if there is one).

(f) The critical region is the range of sample values that leads to the rejection of the null hypothesis.

True The critical region is the range of sample values that leads to the rejection of the null hypothesis.

2. HT Consider a situation in which a restaurant chain will test for arsenic levels in a sample of chickens from a supplier. If there is evidence that the average level of arsenic is over $80\ ppb$, the chain will permanently cancel its relationship with the supplier.

(a) What are the **underlying hypothesis**? | $H_0: \mu \le 80$ | $H_1: \mu > 80$ → Chicken bad! → Relationship will be canceled.

(b) What would it mean for analysts at the restaurant chain to make a *Type I error* in the context of this situation?

A **Type I error** would mean that the restaurant chain incorrectly concluded that the average level of arsenic is over 80 ppb when it is not. (**False Positive** → We **reject the null hypothesis** when it is actually **true**).

(c) What would it mean to make a Type II error in this situation?

A **Type II error** would mean that the restaurant chain incorrectly concluded that the average level of arsenic is not over 80 ppb when it is. (**False Negative** → We **fail to reject the null hypothesis** when it is actually **False**).

2. HT If the **power of a statistical test is increased**, for example by increasing the sample size, how does the probability of a Type II error change? If the power of a statistical test is increased (e.g., by increasing the sample size), the probability of a Type II error **decreases**. This means that the test is more likely to correctly reject the null hypothesis when it is **false**.

Generally, increasing the sample size leads to a more precise estimate of the population parameter, reducing the standard error of the estimate. This, in turn, increases the power of the test, making it more likely to detect an effect if it exists. As a result, the probability of a Type II error decreases.

2. HT A study was conducted by a retail store to determine if the majority of their customers were teenagers. With $\hat{p} = 0.48$, the null hypothesis was not rejected and the company concluded that they did not have evidence that the majority of their customers were teenagers. But, in reality, the proportion of all of their customers (i.e., the population) who are teenagers is actually $p = 0.53$. Did this research study result in a Type I or II error, or correct decision?

In this study: p → Population proportion of teenage customers.

Hypotheses: **Null hypothesis** → $H_0: p \le 0.5$ → "The proportion of teenage customers $p \le 0.5$" (no majority).

Alternative hypothesis → $H_1: p > 0.5$ → "The proportion of teenage customers $p > 0.5$" (majority).

Since $\hat{p} = 0.48$ and $p = 0.53$, the sample proportion is less than the population proportion, indicating that the store failed to reject the null hypothesis when it was actually **incorrect/false**. | Therefore, by failing to reject a false null hypothesis, this study made a Type II error.

2024

1. A: The Poisson distribution is a discrete distribution that can be used to model the time between events occurring at a constant average rate. **FALSE**

B: The probability density function describes the probability function of a continuous variable whereas a probability mass function describe the probability function of a discrete random variable. **TRUE**

+: Probability mass function is used for **continuous values** and Probability density function is used for **discrete values**. **FALSE** [Ao Contrário]

2. Consider an exam that contains 20 questions, each one with four different options available, among which just one is correct. To pass the exam, at least 10 questions must be answered correctly. What kind of distribution is appropriate to measure the probability of an unprepared student to pass the exam? **Binomial Distribution**

3. A: Rejecting the null hypothesis when the null hypothesis is true is a type II error. **FALSE**

B: When the sample size increases, the probability of correctly rejecting the null hypothesis, given that the null hypothesis is false, also increases. **TRUE**

4. Consider the vector heights that contains information regarding the height (in cm) of 100 NOVA IMS students. Considering the **R** outputs below, select the correct option.

```
## mean     lwr.ci   upr.ci
## 168.9654 167.6500 170.2807
```

If we repeat the sampling many times, we say that in 95% of the times the real average height of all the students of NOVA IMS is inside the confidence interval.

5. A: With observational data (i.e., non-experimental data), the simple linear regression model will always uncover the ceteris *paribus* relationship between two variables y and x. **FALSE**

B: Econometric methods can simulate a *ceteris paribus* experiment. **TRUE**

6. Random sampling means that $\{(y_i, x_i)\}$ is an independent sequence. $Cov(y_i, y_j) = Cov(x_i, x_j) = Cov(y_i, x_j) = 0, for\ i \neq j$

7. Suppose the population model is given by: $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i$ (1)

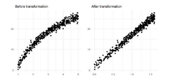
However, you have estimated the following model: $y_i = \beta_0 + \beta_1 x_{i1} + v_i$, where $v_i = \beta_2 x_{i2} + u_i$ (2)
- The OLS estimator of the equation (2) could be biased, if $Cov(x_{i1}, x_{i2}) \neq 0$

8. Consider the model: $log(wage) = \beta_0 + \beta_1\, male + \beta_2\, female + \beta_3\, educ$

Which assumption are violating with this specification? Absence of perfect collinearity

9. Consider the plots below:

After transformation, we have a linear relationship, and the specified model should be $log(y) = \beta_0 + \beta_1 x + u$, so: if x increases by one unit, it is expected that y will increase by $\beta_1 \times 100\%$.



10. Suppose the median starting salary for new law school graduates is determined by:

$$log(salary) = 8.343 + 0.005LSAT + 0.248GPA + 0.095log(libvol) + 0.038log(cost) - 0.003 * rank$$

where $LSAT$ is the median LSAT score for the graduating class, GPA is the median college GPA for the class, $libvol$ is the nº of volumes in the law school library, $cost$ is the annual cost of attending law school (in thousands of $), and $rank$ is a law school ranking (with $rank = 1$ being the best).
- When the annual cost of attending law school increases by 1 thousand dollars, it is expected that the salary will increase by 3.80%, holding all other factors constant.

11. Consider the following model that explains MLB salary (in $):

$$MLB = \beta_0 + \beta_1\, years + \beta_2\, gamesyr + \beta_3\, hruyr + u$$

where *salary* is the total salary in dollars, *years* is the number of years in the league, *gamesyr* is average games played per year, and *hruyr* is ....

A: To test the null hyp. $H_0: \beta_2 + \beta_3 = 0$, it is necessary to estimate the model $MLB = \beta_0 + \beta_1 years + u$. **FALSE**

B: To test for the presence of heteroskedasticity using the **Breusch-Pagan Test**, it is necessary to estimate the auxiliary regression $\hat{u}^2 = \alpha_0 + \alpha_1$ **years** + α_2 **gamesyr** + α_3 home runs + e, where u are the residuals of OLS estimation of equation. **TRUE**

12. Consider the model to estimate the effect of age (age) and annual family income (inc) on net financial wealth ($nettfa$), specified below:

$$nettfa = \beta_0 + \beta_1 inc + \beta_2 age + u$$

- The estimated model is globally significant.
  - $F - statistic$ test the global significance of the model. Since the **p-value of the F-test is less than 2.2e-16** we can state that the model is globally significant.
  - p-value for $age < 2e-16$, which is much smaller than **0.01**. Thus, it is significant at the 1% level.
  - 16.91% of the variation in $nettfa$ is explained by the 2 variables.
  - ~~For the test $H_0: \beta_2 = 1$ against $H_1: \beta_2 < 1$, we reject the null hypothesis at a 5% significance level.~~

    The test would be $H_0: \beta_2 = 1$ against $H_1: \beta_2 \neq 1$.

    We would need to check if 1 is in the interval $[\hat\beta_2 - t_{crit}SE(\hat\beta_2), \hat\beta_2 + t_{crit}SE(\hat\beta_2)]$.

    This is also wrong, we reject that it is different than one, **we can't prove that it is lower**.

```
## Call:
## lm(formula = nettfa ~ inc + age, data = k401ksubs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -509.27 -18.71  -4.09   10.02 1464.74
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -60.69654    2.59633  -23.38   <2e-16 ***
## inc           0.95336    0.02528   37.72   <2e-16 ***
## age           1.03078    0.05912   17.43   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 58.31 on 9272 degrees of freedom
## Multiple R-squared:  0.1691, Adjusted R-squared:  0.1689
## F-statistic: 943.2 on 2 and 9272 DF,  p-value: < 2.2e-16
```

13. RESET test

```
reg <- lm(wage ~ educ + exper + tenure)
auxiliary_reg <- lm(wage ~ educ + exper + tenure + fitted(reg)^2 + fitted(reg)^3)
library(car)
linearHypothesis(reg, c("fitted(reg)^2", "fitted(reg)^3"))
```

14. `reg <- lm(wage ~ educ + exper + tenure, data=wage1)` | `bptest(reg)` `## BP = 43.096, df = 3, p-value = 2.349e-09`

- We are performing the **BP test** and we *reject the null hypothesis* for a 5% significance level, which means we have statistical evidence of the presence of heteroskedasticity. In this setting, the OLS estimator is **not BLUE**.

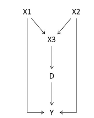
15. (a) If we **reject the null hypothesis** in the Hausman test for panel data, then we can use ~~pooled OLS~~. **FALSE**

(b) Random effects estimation is appropriate when c_i influences the dependent variable but is not correlated with the independent variables. **TRUE**

(c) When c_i may be correlated with the indep. variables, we should use the ~~random effects estimation~~. **FALSE**

16. Consider the following **DAG**: Knowing that the **causal paths** is $D \rightarrow Y$.

Which of the following is a backdoor path? $\mathbf{Y \leftarrow X_1 \rightarrow X_3 \rightarrow D}$



17. Suppose you have estimated the following model: $y_i = \beta_0 + \beta_1 x_{i1} + u_i$

However, you suspect that x_{i1} is endogenous and aim to select an ***instrumental variable***. Considering these candidate variables:

$Cov(z_{i1}, x_{i1}) = 0$ $Cov(z_{i1}, y_i) \neq 0$ $\mathbf{Cov(z_{i2}, x_{i1}) \neq 0}$ $\mathbf{Cov(z_{i2}, y_i) = 0}$ $Cov(z_{i3}, x_{i1}) = 0$ $Cov(z_{i3}, y_i) = 0$ $Cov(z_{i4}, x_{i1}) \neq 0$ $Cov(z_{i4}, y_i) \neq 0$ Which one is appropriate to choose? Z_2

18. We intend to analyze the difference in test scores among undergrad students that were subjected to a change in policy implementation (expansion). We collected the test points for those students before and after the policy implementation. A regression model is estimated below:

$$scores = \beta_0 + \beta_1\, post + \beta_2 treat + \beta_3\, (treat * post) + u$$

What is the value of β_3?

The *intercept* is 38.663, this is the average value of the score when *treat* and ***post*** **= 0.**

The coefficient for *treat* is 6.077, and means the change in *scores* when ***treat*** **= 1**, but only when ***post*****=0**.

The coefficient for *post* is 11.523, and means the change in *scores* when ***post*****=1**, but only when ***treat*****=0**.

β_3 is the interaction term between the **two dummies**. From the provided table, we can calculate the interaction term as follows:

When ***treat*** **= 1 and** ***post*** **= 1, the average score is 66.3.** Using the equation: $66.3 = 38.663 + 6.077 + 11.523 + \beta_3$.

Solving : $\beta_3 = 66.3 - 38.663 - 6.077 - 11.523 = 10.037$

```
data %>%
group_by(treat, post) %>%
summarise(ybar = mean(score))

## # A tibble: 4 x 3
## # Groups:   treat [2]
##    treat  post  ybar
##    <dbl> <dbl> <dbl>
## 1     0     0  38.7
## 2     0     1  50.2
## 3     1     0  44.7
## 4     1     1  66.3
```

19. Suppose you want to model the *growth* in **different countries** using the following regressors: Test whether the estimates associated with *yearsschool*, *rev_coups*, *assassinations*, and *rgdp60* are jointly significant **F-test**.

We need to compute the F-statistic in order to determine whether we can reject the null hypothesis.

The null hyp. is that **all coefficients are jointly 0**: H_0: $\beta_{yearsschool} = \beta_{rev_coups} = \beta_{assasinations} = \beta_{rgdp60} = 0$

The **restricted model** must be fitted without this regressors. Comparing models (1) and (3) we see that: model (1) has $R^2 = 0.291$ with 6 regressors (including the intercept) and model (3) has $R^2 = 0.045$ with 2 regressors (including the intercept). $n = 64$. The degrees of freedom of the **restricted model** is $df_r = 64 - 2 = 62$ and the **unrestricted** is $df_{ur} = 64 - 6 = 58$. $k_{ur} = 6, k_r = 2$

$$F = \frac{(0.291 - 0.045)/(6-2)}{(1-0.291)/(64-6)} = \frac{0.246/4}{0.709/58} \approx 5.03$$

We need to calculate the critical value of the $F\ distribution$ with 4 and 58 ***df***. We can approximate the critical value in the **F distribution** with 4 and 60 degrees of freedom at **5%** as 2.53.

∴Given the high **F statistic** (5.03) **is much higher than the critical value 2.53** we can reject the null hypothesis that **the coefficients are jointly zero.** We conclude that those regressors are jointly significant.

Dependent variable	growth		
	(1)	(2)	(3)
tradeshare	1.341 (0.960)	1.898 (0.936)	1.681 (0.987)
yearsschool	0.564 (0.143)	0.243 (0.084)	
rev_coups	−2.150 (1.119)		
assassinations	0.323 (0.488)		
rgdp60	−0.0005 (0.0002)		
Constant	−0.627 (0.783)	−0.122 (0.663)	0.957 (0.580)
Observations	64	64	64
R²	0.291	0.161	0.045